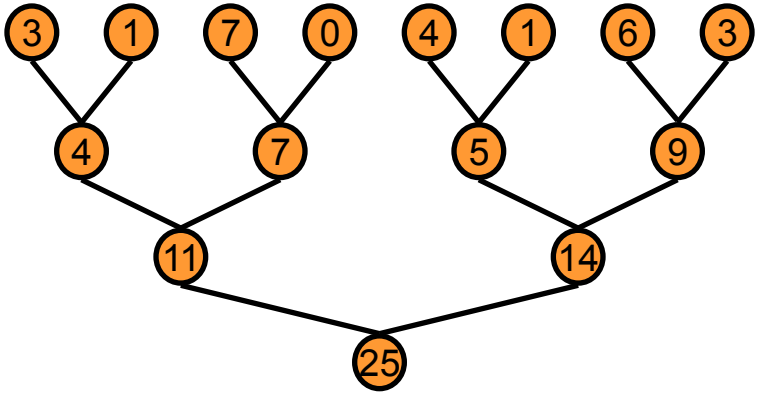
Parallel Reduction

- Tree-based approach used within each thread block



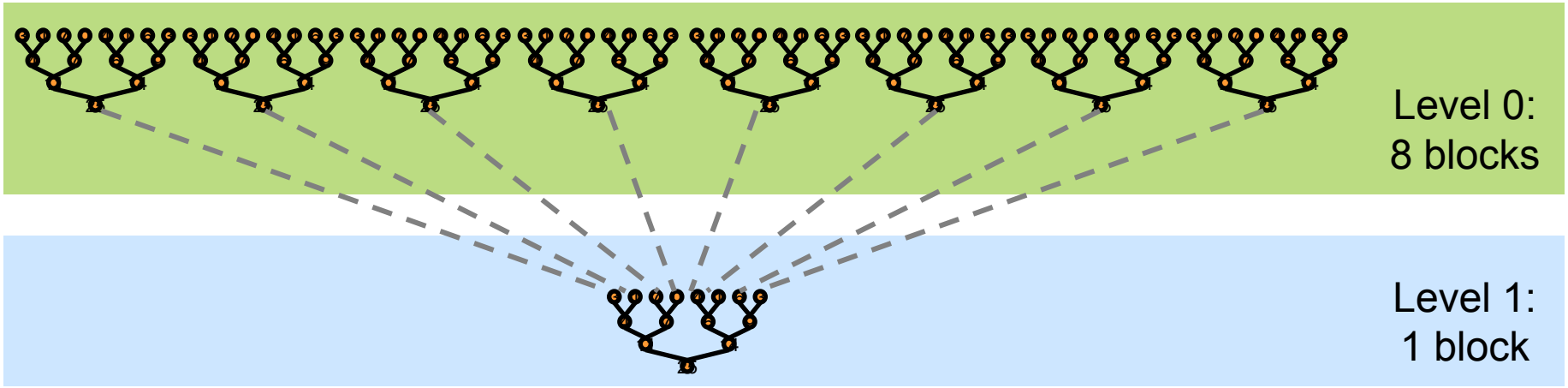
- Need to be able to use multiple thread blocks
  - To process very large arrays
  - To keep all multiprocessors on the GPU busy
  - Each thread block reduces a portion of the array
- But how do we communicate partial results between thread blocks?

Problem: Global Synchronization

- If we could synchronize across all thread blocks, could easily reduce very large arrays, right?
  - Global sync after each block produces its result
  - Once all blocks reach sync, continue recursively
- But CUDA has no global synchronization. Why?
  - Expensive to build in hardware for GPUs with high processor count
  - Would force programmer to run fewer blocks (no more than # multiprocessors * # resident blocks / multiprocessor) to avoid deadlock, which may reduce overall efficiency
- Solution: decompose into multiple kernels
  - Kernel launch serves as a global synchronization point
  - Kernel launch has negligible HW overhead, low SW overhead

Solution: Kernel Decomposition

- **Avoid global sync by decomposing computation into multiple kernel invocations**

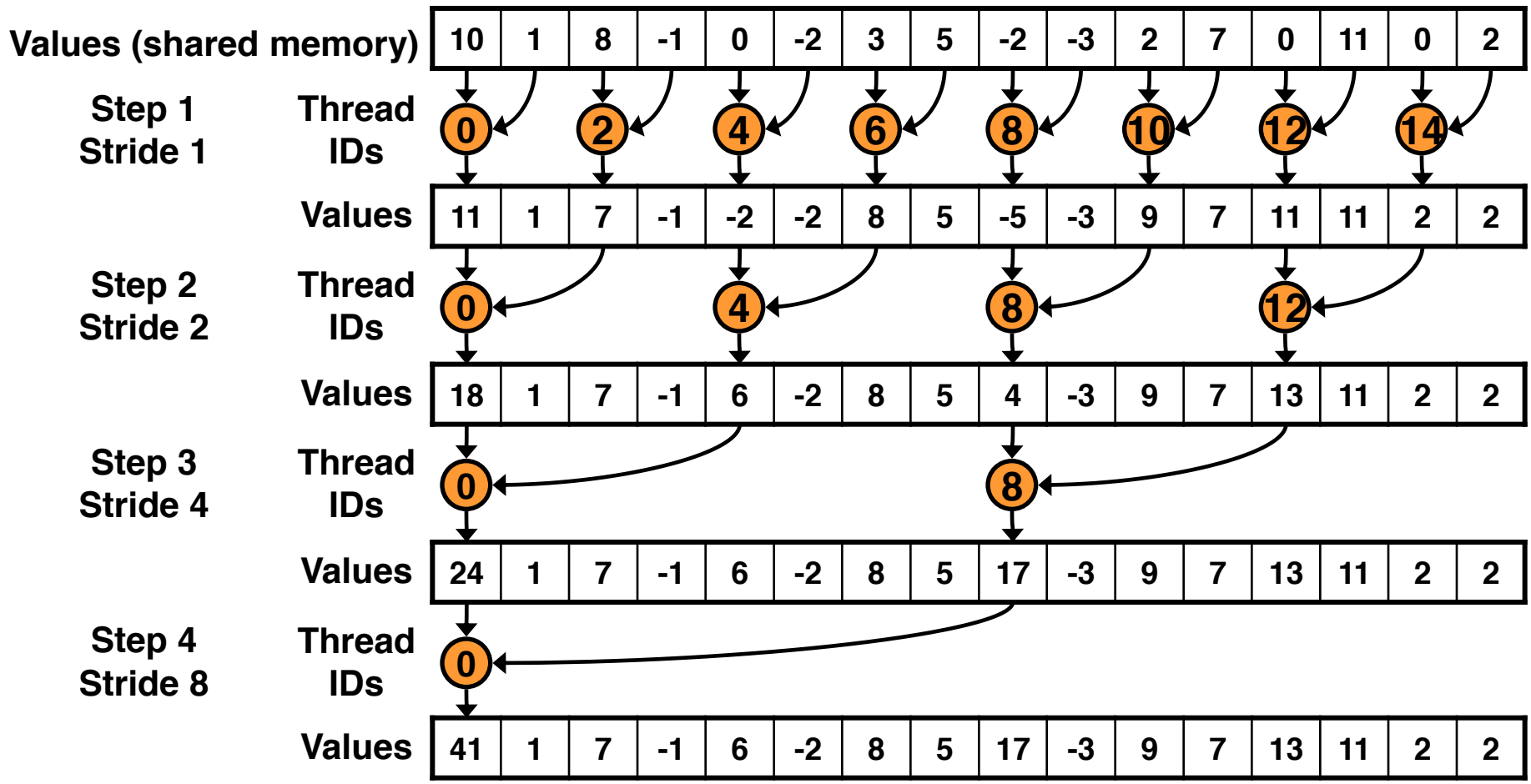


- **In the case of reductions, code for all levels is the same**
  - **Recursive kernel invocation**

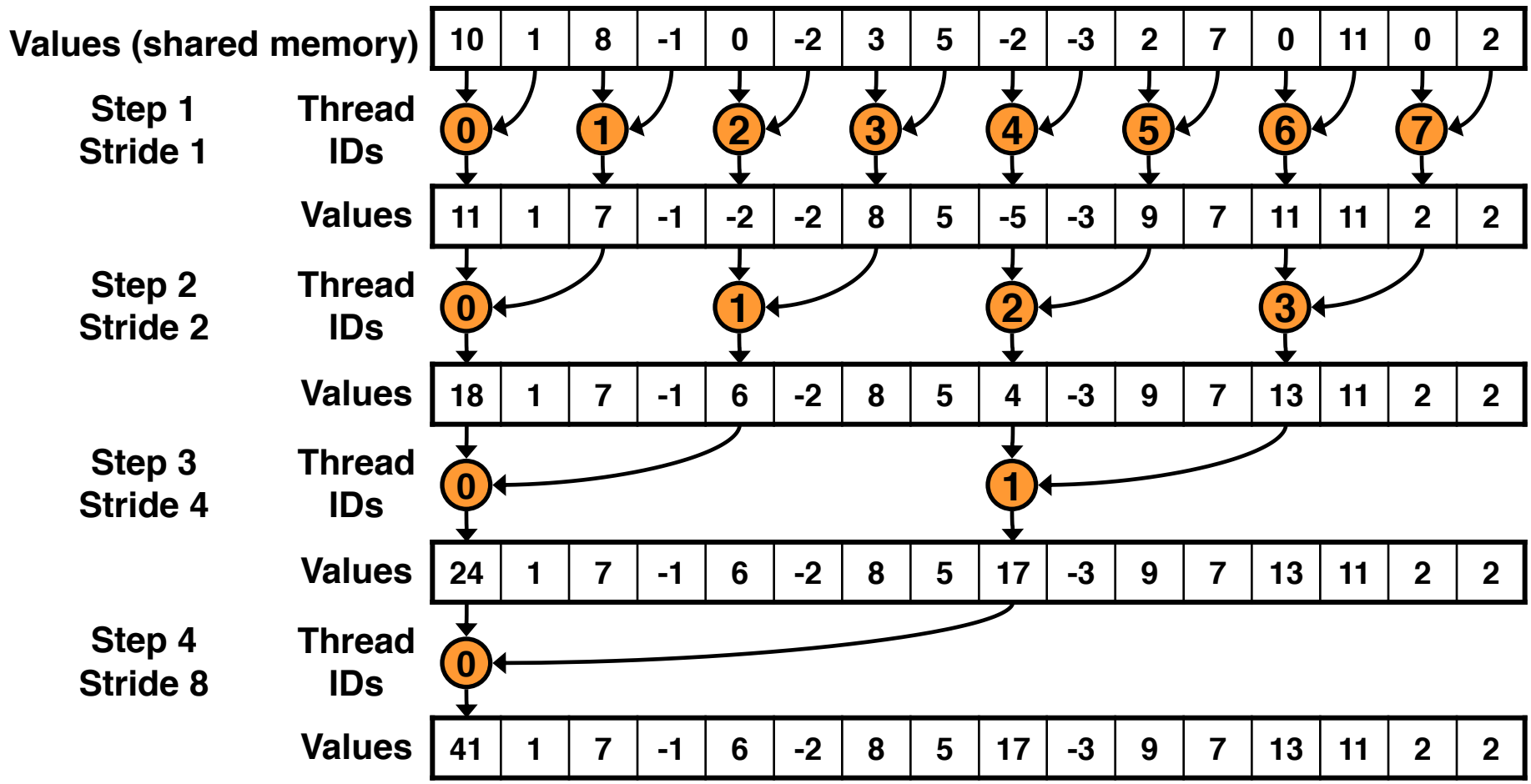
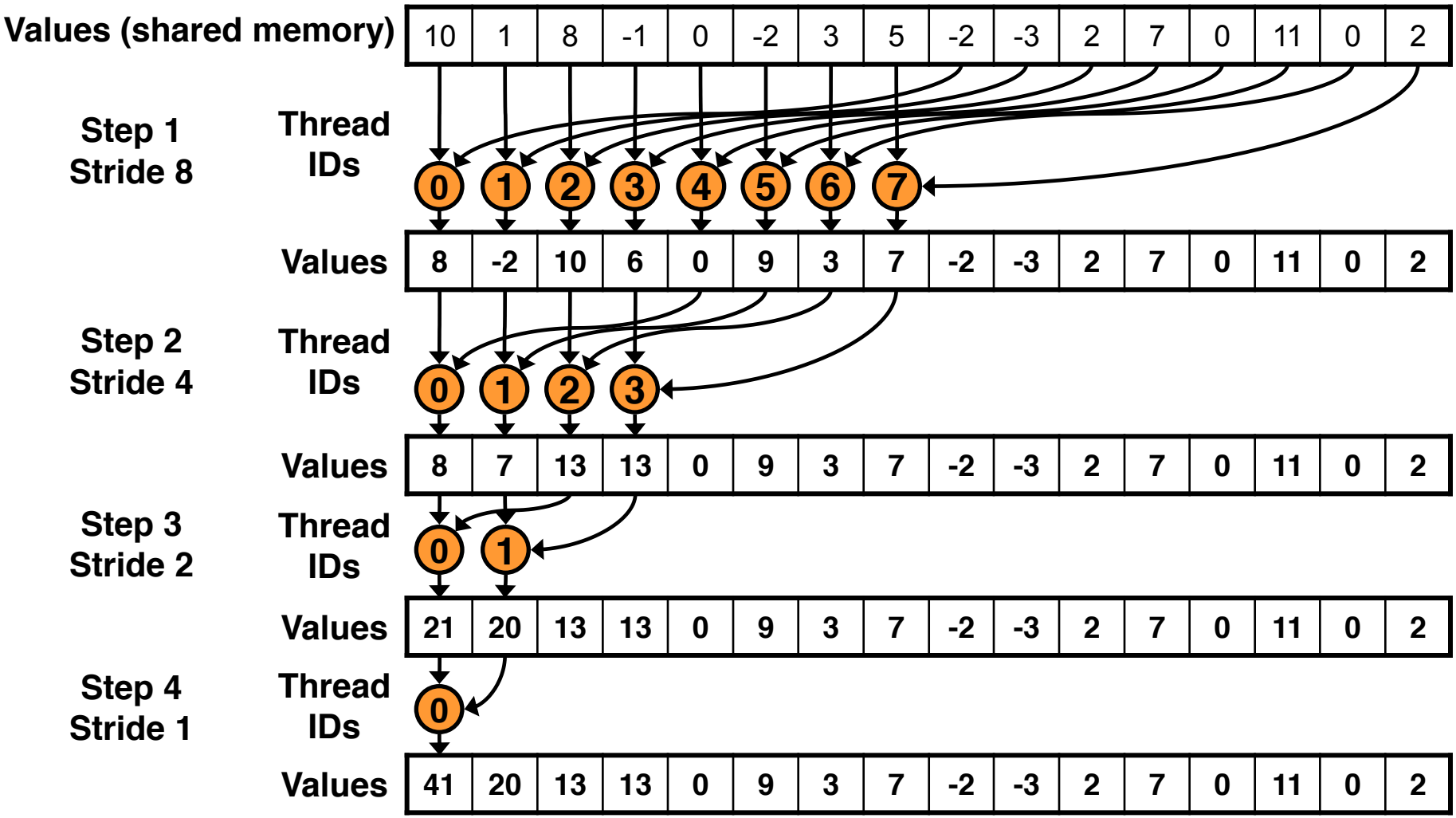
Reduction #1: Interleaved Addressing

```
__global__ void reduce0(int *g_idata, int *g_odata) {
   extern __shared__ int sdata[];

   // each thread loads one element from global to shared mem
   unsigned int tid = threadIdx.x;
   unsigned int i = blockIdx.x*blockDim.x + threadIdx.x;
   sdata[tid] = g_idata[i];
   __syncthreads();

   // do reduction in shared mem
   for(unsigned int s=1; s < blockDim.x; s *= 2) {
      if (tid % (2*s) == 0) {
         sdata[tid] += sdata[tid + s];
      }
      __syncthreads();
   }

   // write result for this block to global mem
   if (tid == 0) g_odata[blockIdx.x] = sdata[0];
}
```

Parallel Reduction: Interleaved Addressing



Parallel Reduction: Interleaved Addressing



Parallel Reduction: Sequential Addressing